University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

STA 2101: Statistics & Probability

Lecture 02:
Sampling Frequency Chart

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Class Code
sdb2r4mu

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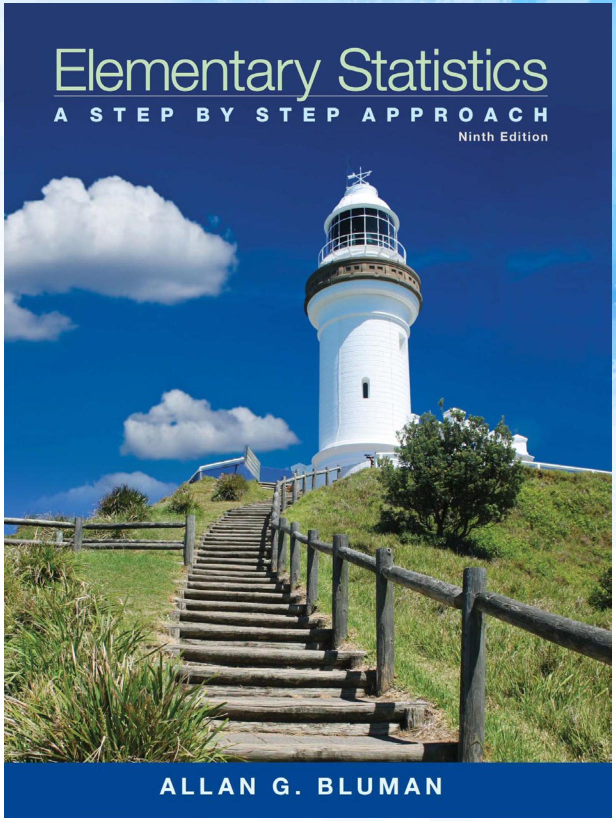
Reference Book

- **Text Book(s)**:
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

- **Reference Material/Book(s)**:
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	10
Project	20
Total	100

In Previous Episode

Statistics & Probability

Previous Learning

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

Statistics is the theory and method of analyzing quantitative data obtained from samples of observations in order to study and compare sources of variance of phenomena, to help make decisions to accept or reject hypothesized relations between the phenomena, and to aid in drawing reliable inferences from empirical observations.

The FBI reported that violent crimes were down by 6.4% in 2011.

- USA TODAY reported that the average college graduate student loan debt was about $19,000.
- The College Stress and Mental Illness Poll reported that 85% of college and university students reported feeling stress daily; 77% reported stress from school work, and 74% experienced stress from grades.
- The Occupational Outlook Handbook reported that the median hourly wage for a registered nurse was $31.10 per hour.
- Reader’s Digest reported that the average cost of using a plasma television is $0.1152 per hour.
- In 2013, the number of sales of smartphones is estimated to be 832.5 million units globally.

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Previous Learning

- **Artificial Intelligence (AI) & Neural Networks:** Statistical tests are used for AI model validation and performance analysis. Concepts like cross-validation, regularization, and gradient-based optimization are fundamental to building and evaluating machine learning models.
- **Bioinformatics:** Statistics is crucial for analyzing biological data, including identifying disease markers and predicting protein structures.
- **Natural Language Processing (NLP) & Computer Vision:** Statistical models are widely used in NLP for tasks like text prediction and information retrieval. In computer vision, they are vital for image processing tasks such as feature extraction and dimensionality reduction using techniques like **Principal Component Analysis (PCA)** and **Singular Value Decomposition (SVD)**.

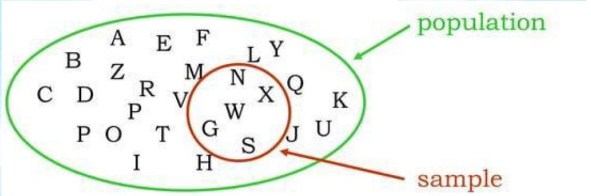
•**Data Mining & Data-Driven Decisions:** Statistics helps find useful patterns in data, enabling businesses to make informed, data-driven decisions. Methods like regression and clustering are key to this process.

•**Data Summarization:** Statistical concepts are essential for summarizing and visualizing data through charts, graphs, and dashboards. The **Pearson Correlation Coefficient**, for example, measures the relationship between variables.

•**Databases:** Statistical models are used to analyze database performance, detect outliers, and optimize data storage using techniques like **Chi-Square Tests** and correlation analysis.

Previous Learning

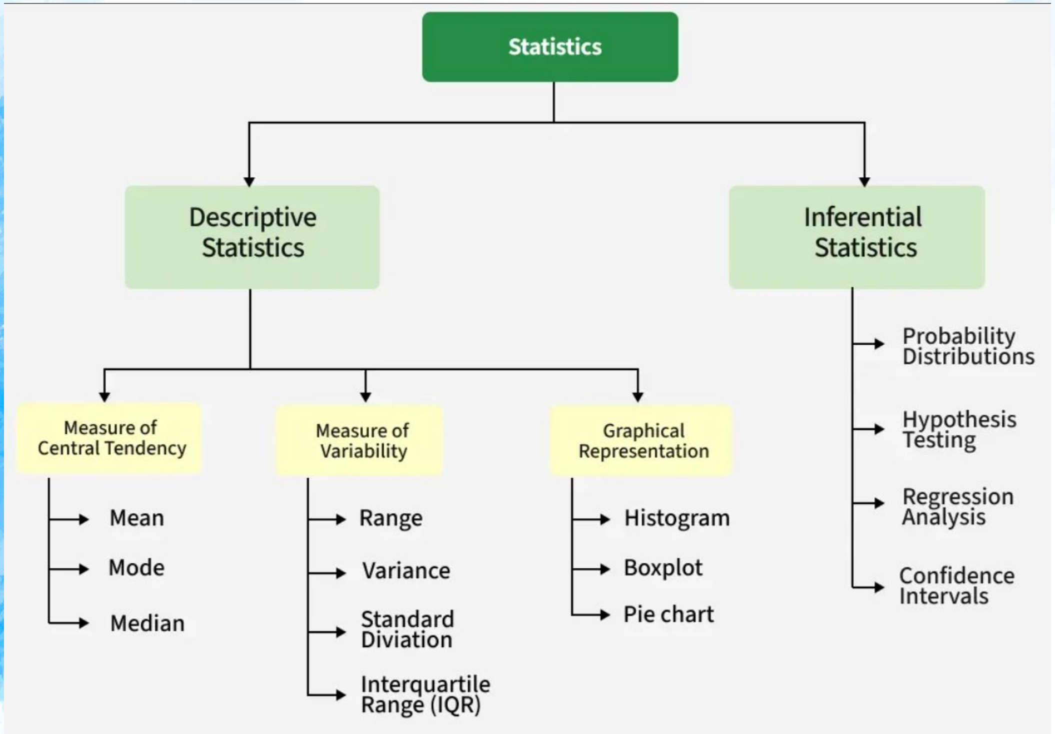
Variables and Data
In statistics, we use **variables** to describe and understand seemingly random situations. A variable is a characteristic or attribute that can take on different values. The actual measurements or observations a variable can assume are called **data**. A collection of these data values is known as a **data set**, and each individual value within it is a **data value** or **datum**. When a variable's values are determined by chance, it's called a **random variable**.



Variable: A variable is a characteristic or attribute that can assume different values. Data are the values (measurements or observations) that the variables can assume.

Population and sample: A population is the entire group that you want to draw conclusions about. **A sample** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.

Previous Learning



- Statistics is sometimes **divided into two main areas**, depending on how data are used. The two areas are:
- 1. Descriptive statistics
- 2. Inferential statistics

- **1. Descriptive statistics:** Descriptive statistics consists of the collection, organization, summarization, and presentation of data. In descriptive statistics the statistician tries to describe a situation.

- **2. Inferential statistics:** Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.

Previous Learning

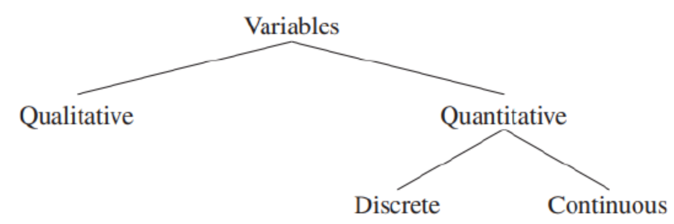
Variables are characteristics or attributes that can be classified into two main types: **qualitative** and **quantitative**.

Qualitative Variables
Qualitative variables represent categories or descriptions and cannot be counted or measured numerically.
Example: Gender (male, female, other) or hair color (black, brown, blonde).

Quantitative Variables
Quantitative variables are numerical and can be counted or measured.
Example: Age, height, or weight.
Quantitative variables are further divided into two subgroups:

- **Discrete Variables:** These variables can be counted and assume specific, distinct values. They typically represent whole numbers.
  - **Example:** The number of students in a class or the number of cars in a parking lot.
- **Continuous Variables:** These variables can assume any value within a given range, including fractions and decimals. They are obtained by measuring.
  - **Example:** A person's height (e.g., 175.5 cm) or the temperature of a room (e.g., 22.8°C)

The classification of variables can be summarized as follows:



Previous Learning

Class boundaries represent the range of values that a rounded data point could fall within before rounding.
For example, a recorded height of 73 inches could actually be anywhere between 72.5 and 73.5 inches. Boundaries are written for convenience, like 72.5–73.5, but mean all values up to, but not including, the upper limit.

EXAMPLE 1–3 Class Boundaries

Find the boundaries of each variable.

a. 8.4 quarts
b. 138 mmHg
c. 137.63 mg/dL

SOLUTION

a. 8.35–8.45 quarts
b. 137.5–138.5 mmHg
c. 137.625–137.635 mg/dL

TABLE 1–1 Recorded Values and Boundaries

Variable	Recorded value	Boundaries
Length	15 centimeters (cm)	14.5–15.5 cm
Temperature	86 degrees Fahrenheit (°F)	85.5–86.5°F
Time	0.43 second (sec)	0.425–0.435 sec
Mass	1.6 grams (g)	1.55–1.65 g

Today's Topics

- Sampling Methods

Sampling Methods

- **Types of Sampling**

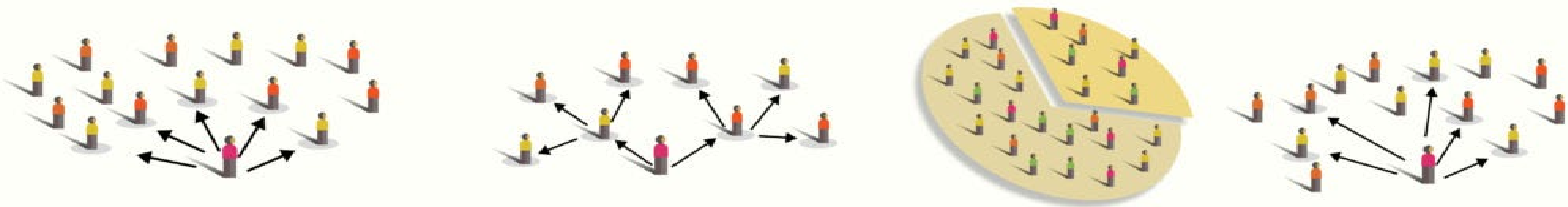
1. **Probability Sampling** (each member has a known chance of selection)
   - **Simple Random Sampling**: equal chance for all
   - **Systematic Sampling**: selecting every kth element
   - **Stratified Sampling**: dividing into groups (strata) and sampling from each
   - **Cluster Sampling**: dividing into clusters, then randomly selecting clusters

2. **Non-Probability Sampling** (not every member has a chance of selection)
   - **Convenience Sampling**: based on ease of access
   - **Judgment/Purposive Sampling**: based on researcher's judgment
   - **Quota Sampling**: ensuring representation of certain groups
   - **Snowball Sampling**: participants recruit other participants

Probability Sample Methods

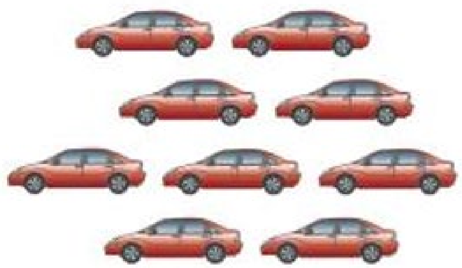
Simple Random Sampling | Stratified Random Sampling | Cluster Sampling | Systematic Sampling

Non Probability Sample Methods



Convenience Sampling | Snowball Sampling | Quota Sampling | Purposive or Judgmental Sampling

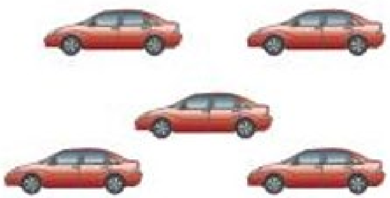
1. Random



Population

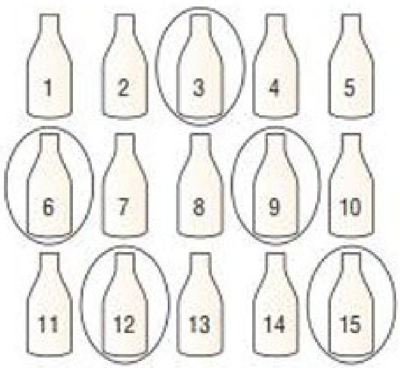
Table D Random Numbers

10480	15011	01536	02011	81647
22368	46573	25595	85393	30995
24130	48360	22527	97265	76393
42167	93093	06243	61680	07856
37570	39975	81837	16656	06121
77921	06907	11008	42751	27750
99562	72905	56420	69994	98872
96301	91977	05463	07972	18876
89579	14342	63661	10281	17453
85475	36857	43342	53988	
28918	69578	88321		
63553	40961			

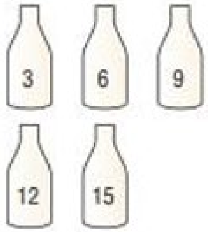


Sample

2. Systematic

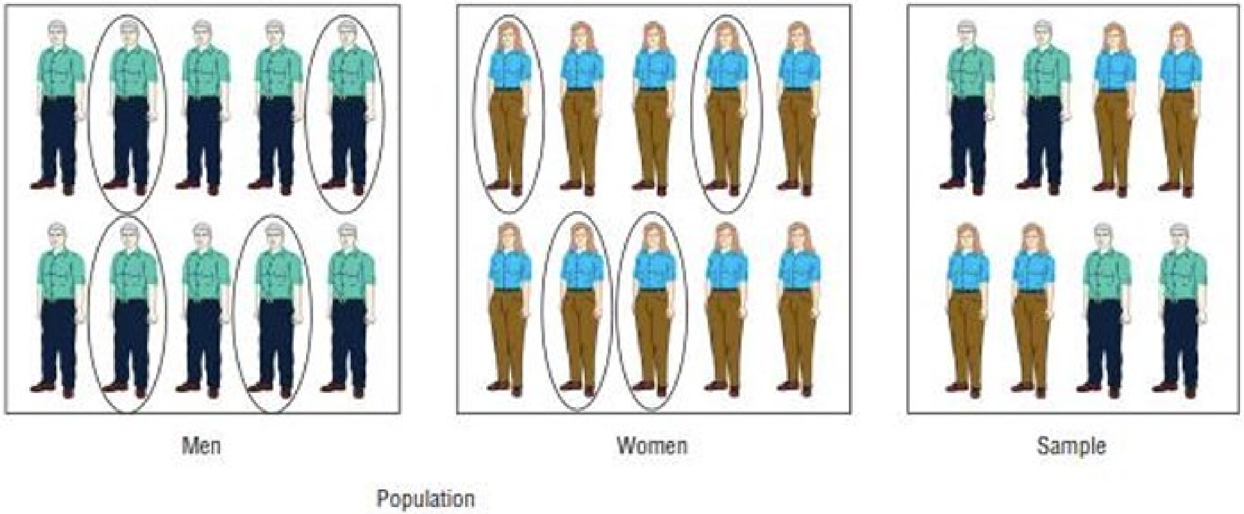


Population



Sample

3. Stratified



4. Cluster

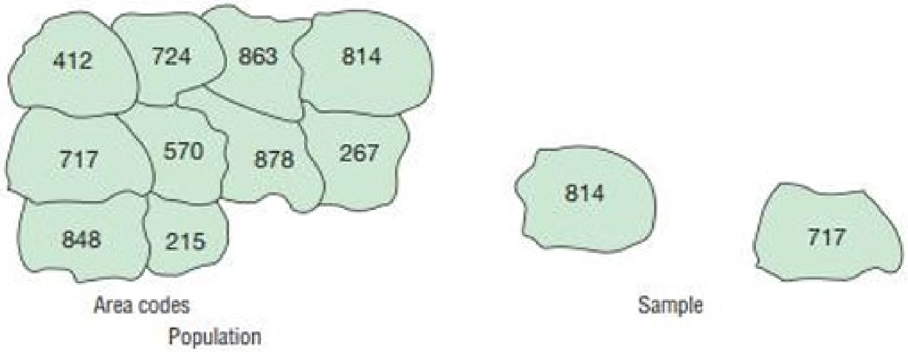


TABLE 1–4 Summary of Sampling Methods	
Random	Subjects are selected by random numbers.
Systematic	Subjects are selected by using every *k*th number after the first subject is randomly selected from 1 through *k*.
Stratified	Subjects are selected by dividing up the population into subgroups (strata), and subjects are randomly selected within subgroups.
Cluster	Subjects are selected by using an intact subgroup that is representative of the population.

Example Practice

EXAMPLE 1–5 Sampling Methods

State which sampling method was used.

a. Out of 10 hospitals in a municipality, a researcher selects one and collects records for a 24-hour period on the types of emergencies that were treated there.

b. A researcher divides a group of students according to gender, major field, and low, average, and high grade point average. Then she randomly selects six students from each group to answer questions in a survey.

c. The subscribers to a magazine are numbered. Then a sample of these people is selected using random numbers.

d. Every 10th bottle of Super-Duper Cola is selected, and the amount of liquid in the bottle is measured. The purpose is to see if the machines that fill the bottles are working properly.

SOLUTION

a. Cluster

b. Stratified

c. Random

d. Systematic

Thank You

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